

COMPUTER NETWORKS – HANDWRITTEN EXAM NOTES

UNIT 2: Data Link Layer, Framing, Flow & Error Control, CRC, Checksum, Sliding Window, Hamming Code

B.Tech Mid-1 Examination Preparation | Complete Question Bank Solutions

PART A – 2 MARKS QUESTIONS

Question 1

What is framing in the data link layer?

Definition & Purpose

Framing is a fundamental Data Link Layer technique in which the continuous raw bit stream received from the physical layer is divided into manageable, distinct data units called **Frames**. Each frame consists of a **Header** (addressing and control info), **Payload** (network layer packet), and **Trailer** (error detection bits / FCS).

Importance

- Allows the receiver to identify where a message starts and ends (synchronization).
- Enables individual error detection and flow control per frame.

Question 2

Define flow control

Definition

Flow Control is a mechanism in the data link layer and transport layer that coordinates the data transmission rate between a fast sender and a slower receiver to prevent the receiver's memory buffer from overflowing and dropping frames.

Approaches

- **Feedback-based Flow Control:** The sender sends frames only after receiving permission (ACK / Window update) from the receiver (e.g., Stop-and-Wait, Sliding Window).
- **Rate-based Flow Control:** Built-in transmission rate limits without explicit frame-by-frame acknowledgment.

Question 3

What is CRC?

Definition & Properties

CRC (Cyclic Redundancy Check) is a powerful polynomial division-based error detection method operating on binary modulo-2 arithmetic (XOR logic). The sender appends a redundant sequence of r bits (the remainder of dividing the message by a predefined generator polynomial $G(x)$) to the data stream.

Key Capabilities

- Detects 100% of all single-bit errors and double-bit errors.
- Detects 100% of all odd numbers of bit errors.
- Detects all burst errors of length $\leq r$ bits.
- Widely used in Ethernet (CRC-32) and Wi-Fi frames.

Question 4

What is the purpose of a checksum in error detection?

Purpose & Mechanism

The Checksum is an error-detection technique used predominantly in higher layer protocols (TCP, UDP, IPv4) to verify that data has not been corrupted during transit.

- **Sender Side:** The data is treated as a sequence of 16-bit integers. They are added together using 1's complement arithmetic (adding any carry bit back to the LSB). The 1's complement of this sum is calculated and appended as the checksum field.
- **Receiver Side:** The receiver sums all incoming words including the checksum. If the result contains all 1s (0xFFFF), the data is error-free; otherwise, an error occurred and the packet is discarded.
- **Advantage:** Extremely fast and simple to implement in software compared to complex hardware division.

Question 5

Explain the basic idea of the Stop-and-Wait protocol

Basic Working Idea

In the Stop-and-Wait Protocol (a 1-bit sliding window protocol with window size $W_s = 1$, $W_r = 1$):

1. The sender transmits **one data frame** and stops, starting a retransmission timer.
2. The sender waits until it receives a positive acknowledgment (**ACK**) from the receiver before sending the next frame.
3. If the frame or ACK is lost in transit and the timer expires (**Timeout**), the sender retransmits the same frame.
4. **Sequence Numbers (0 and 1)**: Alternating 1-bit sequence numbers prevent the receiver from accepting duplicate frames if an ACK was delayed.

----- PART B – 5 MARKS / ESSAY QUESTIONS -----

Essay Question 1

Explain fixed-size and variable-size framing techniques with suitable examples

1. Introduction

Framing separates the physical bit stream into distinct logical packets. Framing techniques are divided into two broad categories: **Fixed-Size Framing** and **Variable-Size Framing**.

2. Fixed-Size Framing

In fixed-size framing, every frame has an identical, predefined length. There is no need for explicit frame boundary delimiters because the receiver counts the exact number of arriving bits.

- **Example**: ATM (Asynchronous Transfer Mode) networks use fixed 53-byte cells (5 bytes header + 48 bytes payload).
- **Advantage**: Simple hardware implementation and minimal boundary parsing overhead.
- **Disadvantage**: Internal fragmentation when data size does not match frame size (requires zero-padding).

3. Variable-Size Framing Techniques

In variable-size framing, frame length varies dynamically. Boundary detection requires explicit delimiter mechanisms:

(a) Character Count Method

- **Mechanism**: The first byte in the frame header specifies the total number of characters in the frame (including the count byte itself).
- **Example**: [5][D][A][T][A] represents a 5-character frame.
- **Fatal Vulnerability**: If a single transmission error corrupts the count byte (e.g., 5 becomes 7), the receiver loses synchronization permanently for all subsequent frames! Hence, this method is rarely used alone.

(b) Byte Stuffing (Character-Oriented Framing)

- **Mechanism:** Special 1-byte control characters called **FLAG bytes** (e.g., 0x7E or DLE STX/DLE ETX) mark the start and end of each frame.
- **The Problem:** If binary payload data contains the same bit pattern as the FLAG byte, the receiver will mistakenly terminate the frame prematurely.
- **The Solution (Byte Stuffing):** Whenever the sender encounters a FLAG or Escape (ESC) character in the user data, it inserts an extra escape character (ESC) immediately before it. The receiver automatically removes the ESC byte, restoring the original payload.

(c) Bit Stuffing (Bit-Oriented Framing)

- **Mechanism:** Used in HDLC and modern protocols. A unique 8-bit delimiter flag **01111110** (six consecutive 1s) marks the beginning and end of every frame.
- **Sender Rule (Zero-Insertion):** Whenever the sender encounters **five consecutive 1s (11111)** in the data stream, it automatically inserts a **0** bit immediately after them, regardless of what the next bit is!
- **Receiver Rule (Zero-Deletion):** Whenever the receiver detects five consecutive 1s followed by a 0, it strips the 0 bit. If it sees six consecutive 1s followed by 0 (01111110), it recognizes the true end of the frame!
- **Example:**

Original Data: 0110111111011111001

After Stuffing: 011011111010111110001 (Zeros inserted after every five 1s)

Essay Question 2

Explain error detection and correction techniques used in the data link layer

1. Types of Errors

- **Single-Bit Error:** Only one bit in the entire data unit is flipped from 0 to 1 or 1 to 0.
- **Burst Error:** Two or more consecutive bits are corrupted (common in transmission lines due to electromagnetic noise bursts).

2. Error Detection Techniques

1. **VRC (Vertical Redundancy Check / Simple Parity):** Appends 1 parity bit to ensure the total number of 1s is even (Even Parity) or odd (Odd Parity). Detects all odd numbers of bit errors; fails completely on even-bit errors (e.g., 2 bits flipped).
2. **LRC (Longitudinal Redundancy Check / 2D Parity):** Organizes data as a matrix of rows and columns, calculating parity for both rows and columns. Detects burst errors better than simple VRC.
3. **Checksum:** Sums 16-bit segments using 1's complement arithmetic and sends the complement. Verified at receiver by ensuring sum equals 0xFFFF.

4. **CRC (Cyclic Redundancy Check):** Polynomial division modulo-2. Most powerful detection method with 100% burst detection up to polynomial degree r .

3. Error Correction (Forward Error Correction – FEC)

Error correction enables the receiver not only to detect that an error occurred, but also to identify the exact position of corrupted bits and flip them back without requesting retransmission.

Hamming Code Concept & Minimum Hamming Distance

- **Hamming Distance $d(x,y)$:** The number of bit positions in which two codewords differ.
- **Detection Rule:** To detect up to d bit errors, the minimum Hamming distance of the code must satisfy:

$$d_{\min} \geq d + 1$$

- **Correction Rule:** To correct up to t bit errors, the minimum Hamming distance must satisfy:

$$d_{\min} \geq 2t + 1$$

- **Redundant Bits Formula (Hamming Inequality):** For m data bits and r parity bits:

$$2^r \geq m + r + 1$$

Essay Question 3

Explain CRC and Internet checksum methods used for error detection

1. CRC (Cyclic Redundancy Check)

CRC treats binary strings as polynomials with coefficients in $GF(2)$ (where addition and subtraction are identical to bitwise XOR with no carry).

Algorithmic Steps:

1. Let the data message be $M(x)$ of length k bits and the Generator Polynomial be $G(x)$ of degree r (length $r+1$ bits).
2. **Step 1 (Zero Padding):** Append r zeros to the right of the message: $M(x) \cdot 2^r$.
3. **Step 2 (Modulo-2 Division):** Divide $M(x) \cdot 2^r$ by $G(x)$ using binary XOR division.
4. **Step 3 (Codeword Generation):** Obtain the remainder $R(x)$ of length r bits. The transmitted codeword is:

$$T(x) = M(x) \cdot 2^r + R(x)$$

5. **Receiver Verification:** The receiver divides the received codeword $T(x)$ by the exact same generator $G(x)$. If the remainder is all zeros (000...), the frame is accepted; otherwise, it is rejected!

2. Internet Checksum Method

Used in IP, UDP, and TCP headers (RFC 1071).

Algorithmic Steps:

1. **Division into 16-bit blocks:** The message is partitioned into consecutive 16-bit words. If the message length is odd, one zero byte is padded at the end.
2. **Summation with 1's Complement Addition:** All 16-bit words are added together. Whenever a carry bit overflows past the 16th bit, it is wrapped around and added to the least significant bit (LSB).
3. **1's Complement Inversion:** The bitwise NOT of the final sum is taken to form the Checksum.
4. **Verification:** The receiver sums all incoming words including the checksum. The result must be 0xFFFF (all 1s), which inverted gives 0x0000.

Essay Question 4

Explain Sliding Window protocols with reference to One-bit, Go-Back-N and Selective Repeat protocols

1. Concept of Sliding Window

Sliding Window protocols allow the sender to transmit multiple frames before receiving an acknowledgment, keeping the channel pipelined and maximizing bandwidth utilization.

2. Detailed Protocol Comparison

(a) One-Bit Sliding Window (Stop-and-Wait ARQ)

- **Window Sizes:** Sender Window $W_s = 1$, Receiver Window $W_r = 1$.
- **Sequence Numbers:** Uses 1-bit sequence numbers (0 and 1).
- **Efficiency:** Low channel utilization on high bandwidth-delay product links ($\eta = \frac{1}{1 + 2a}$, where $a = T_p / T_t$).

(b) Go-Back-N (GBN) ARQ

- **Window Sizes:** Sender Window $W_s = 2^m - 1 = N$ (where m is sequence bit length), Receiver Window $W_r = 1$.
- **Mechanism:** The sender can transmit up to N frames continuously without waiting for an ACK.
- **Cumulative ACK:** ACK(k) confirms receipt of all frames up to $k-1$.

- **Error Handling:** If frame k is damaged or lost, the receiver discards frame k and all subsequent arriving frames ($k+1, k+2, \dots$) because $W_r = 1$. When the sender's timer for frame k expires, the sender must **Go Back and Retransmit all N unacknowledged frames!**

(c) Selective Repeat (SR) ARQ

- **Window Sizes:** Sender Window $W_s = 2^{m-1}$, Receiver Window $W_r = 2^{m-1}$ (Sender and receiver windows are equal).
- **Individual ACK & Negative ACK (NAK):** The receiver has internal buffers. When a damaged frame arrives, the receiver sends a **NAK** for that specific frame and buffers out-of-order frames.
- **Error Handling:** The sender retransmits **ONLY** the specific damaged frame, avoiding redundant retransmission of successfully received frames.

3. Comprehensive Comparison Table

Feature	Stop-and-Wait	Go-Back-N (GBN)	Selective Repeat (SR)
Sender Window (W_s)	1	$2^m - 1$ (N)	2^{m-1}
Receiver Window (W_r)	1	1	2^{m-1} ($= W_s$)
Out-of-order Buffering	No	No (Discarded)	Yes (Buffered)
Retransmission	1 frame on timeout	Entire window of N frames	Only corrupted frame
Complexity	Simplest	Moderate	Highest (complex buffer logic)
Bandwidth Efficiency	Lowest	Moderate	Highest

Essay Question 5 (NUMERICAL PROBLEMS)

Problems on CRC, Check sum, Hamming code

Problem 1: Cyclic Redundancy Check (CRC) Calculation

Given: Data Bit Stream $M = 100100$, Generator Polynomial $G(x) = x^3 + x^2 + 1$
 \implies $G = 1101$.

Find the transmitted codeword and verify at receiver.

Step 1: Determine number of zeros to append

Degree of generator $G(x)$ is $r = 3$.

Append $r=3$ zeros to M : $M' = 100100000$.

Step 2: Binary Modulo-2 Division (XOR Division)

```

      1 1 1 1 0 1 (Quotient)
1 1 0 1 | 1 0 0 1 0 0 0 0 0
      1 1 0 1
      -----
        1 0 0 0
        1 1 0 1
        -----
          1 0 1 0
          1 1 0 1
          -----
            1 1 1 0
            1 1 0 1
            -----
              0 1 1 0 (Bring down next bit)
              0 0 0 0
              -----
                1 1 0 0
                1 1 0 1
                -----
                  0 0 1 (CRC Remainder R)
```

Step 3: Transmitted Codeword

Codeword $T = M \cdot 2^r + R = \mathbf{100100001}$

Problem 2: Internet Checksum Calculation

Given: Two 16-bit words:

$\text{Word}_1 = \mathbf{10011001\ 11100010}$

$\text{Word}_2 = \mathbf{00100100\ 10101001}$

Calculate the Checksum and show receiver verification.

Sender Side:

```

  1 0 0 1 1 0 0 1  1 1 1 0 0 0 1 0 (Word 1)
+ 0 0 1 0 0 1 0 0  1 0 1 0 1 0 0 1 (Word 2)
-----
  1 0 1 1 1 1 1 0  1 0 0 0 1 0 1 1 (Sum, no carry)
```


1's Complement (Checksum):

0 1 0 0 0 0 0 1 0 1 1 1 0 1 0 0

Receiver Verification:

1 0 1 1 1 1 1 0 1 0 0 0 1 0 1 1 (Sum of Word 1 + Word 2)
+ 0 1 0 0 0 0 0 1 0 1 1 1 0 1 0 0 (Checksum)

1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 (All 1s → 0xFFFF)

Inverting gives 0x0000 → Data is Accepted (No Error)!

Problem 3: 7-Bit Hamming Code Generation and Single-Bit Error Correction

Given: 4-bit data $D = 1011$ (where $d_7=1, d_6=0, d_5=1, d_3=1$).

Generate 7-bit even parity Hamming Code. If bit position 5 flips during transit, detect and correct it.

Step 1: Bit Position Layout ($n = 7$)

Bit Position	7	6	5	4	3	2	1
Designation	d_7	d_6	d_5	p_4	d_3	p_2	p_1
Value	1	0	1	?	1	?	?

Step 2: Parity Bit Calculations (Even Parity)

- p_1 (checks bits 1, 3, 5, 7): $p_1 \oplus d_3 \oplus d_5 \oplus d_7 = 0 \implies p_1 \oplus 1 \oplus 1 \oplus 1 = 0 \implies p_1 \oplus 1 = 0 \implies p_1 = 1$.
- p_2 (checks bits 2, 3, 6, 7): $p_2 \oplus d_3 \oplus d_6 \oplus d_7 = 0 \implies p_2 \oplus 1 \oplus 0 \oplus 1 = 0 \implies p_2 \oplus 0 = 0 \implies p_2 = 0$.
- p_4 (checks bits 4, 5, 6, 7): $p_4 \oplus d_5 \oplus d_6 \oplus d_7 = 0 \implies p_4 \oplus 1 \oplus 0 \oplus 1 = 0 \implies p_4 \oplus 0 = 0 \implies p_4 = 0$.

Transmitted 7-bit Codeword: 1 0 1 0 1 0 1

Step 3: Error Detection & Correction at Receiver

Suppose bit 5 is corrupted ($1 \rightarrow 0$), so received codeword is 1 0 0 0 1 0 1:

$$\begin{aligned}s_1 &= b_1 \oplus b_3 \oplus b_5 \oplus b_7 = 1 \oplus 1 \oplus 0 \oplus 1 = 1 \\s_2 &= b_2 \oplus b_3 \oplus b_6 \oplus b_7 = 0 \oplus 1 \oplus 0 \oplus 1 = 0 \\s_4 &= b_4 \oplus b_5 \oplus b_6 \oplus b_7 = 0 \oplus 0 \oplus 0 \oplus 1 = 1\end{aligned}$$

Syndrome Word: $S = s_4 s_2 s_1 = 101_2 = 5_{10}$ (Decimal 5)

Conclusion: Bit position 5 is erroneous. Flip bit 5 from 0 to 1 → Restored Codeword: 1010101.

----- LAST-MINUTE EXAM REVISION (UNIT 2) -----

Quick Formula & Protocol Comparison Summary

Concept	Key Formula / Rule	Exam Significance
Bit Stuffing	Insert 0 after five consecutive 1s	Prevents payload data from matching flag 01111110
Hamming Inequality	$2^r \geq m + r + 1$	Calculates minimum parity bits for m data bits
Error Detection Distance	$d_{\min} \geq d + 1$	Minimum distance to detect d errors
Error Correction Distance	$d_{\min} \geq 2t + 1$	Minimum distance to correct t errors
Stop-and-Wait ARQ	$W_s = 1, W_r = 1$	Efficiency $\eta = 1 / (1 + 2a)$
Go-Back-N ARQ	$W_s = 2^m - 1, W_r = 1$	Retransmits all N frames on timeout
Selective Repeat ARQ	$W_s = 2^{m-1}, W_r = 2^{m-1}$	Retransmits only the corrupted frame